

Mortal Computation via Multi-Scale Interoceptive Morphogenesis

A variational, resource-bounded framework for self-modifying embodied intelligence

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GitHub: <https://github.com/Devanik21/The-Invention-Archive>

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Research posture. This manuscript proposes and numerically stress-tests a formal architecture; it is not presented as evidence that artificial systems are alive or conscious. The central object is an explicitly bounded computational agent whose inference, learning, and topology are coupled to an internal viability state. The free-energy quantities used below are variational/information-theoretic unless explicitly identified as physical work or heat. The numerical study verifies consequences of the reduced model rather than establishing biological truth.

Abstract

The dominant paradigm of artificial intelligence treats software state as nearly independent of the continuing integrity of its physical substrate. Biological systems do not enjoy that separation: metabolism, sensing, structure, learning, and repair are coupled inside a finite viability envelope. This paper introduces *Mortal Computation via Multi-Scale Interoceptive Morphogenesis*, a formal design program in which an artificial agent minimizes variational inference error while simultaneously maintaining an internal physiological state and paying explicit costs for activity, plasticity, and topology. The proposed MILLS framework, or *Multi-timescale Interoceptive Learning and Lifespan Selection*, separates fast inference, intermediate parameter adaptation, and slow morphological selection. A relaxed graph mask converts topology into a dynamical state, allowing low-utility connections to be removed when internal reserve is scarce and allowing useful structure to regrow when reserve recovers. A higher-level latent goal modulates lower-level setpoints and therefore enters both local dynamics and structural selection, providing an operational notion of top-down causality without introducing a separate homunculus. The formalism is expressed on a Riemannian parameter manifold, admits a stochastic open-system extension, and can be represented in tensor-network form. An independent SciPy/NumPy simulation integrates the governing reduced equations under increasing environmental disturbance and a structural lesion. At severe noise, the adaptive agent maintains a nonzero viability reserve where the fixed-topology baseline reaches the absorbing zero-reserve state; this improvement is purchased by higher task error and transiently sparse topology. The resulting trade-off is a falsifiable prediction: mortality pressure should induce resource-efficient architectural contraction, but sufficiently aggressive contraction must reduce computational fidelity. The paper therefore frames mortality not as a metaphor but as a measurable control variable that couples information processing to substrate integrity.

1 Introduction

Artificial intelligence has historically benefited from a clean abstraction boundary: an algorithm can be described independently of the hardware that executes it, and hardware can often be replaced without changing the semantics of the program. That abstraction is extraordinarily useful, but it can conceal a design degree of freedom that biology exploits continuously: the computational substrate itself is plastic. Brains change synaptic strengths and structures; bodies remodel; cells regulate internal variables; and living systems spend energy to remain within a viability region. Neural plasticity is not

metabolically free, and structural plasticity can reorganize connectivity in response to homeostatic demands [Li and van Rossum, 2020, Butz and van Ooyen, 2014]. Brain-scale energy budgets further constrain activity and architecture [Herculano-Houzel, 2011].

The proposal to use *mortal computation* as a foundation for biomimetic intelligence already has a substantial theoretical precedent. Ororbias and Friston formulated mortality through Markov blankets and the circular causal relations among inference, learning, and selection [Ororbias and Friston, 2023]. The theme remains active: a 2026 Behavioral and Brain Sciences commentary by Friston explicitly discusses thermodynamic issues attending mortal computation and their implications for conscious artefacts, while Hohwy separately examines mortal computation through self-evidencing [Friston, 2026, Hohwy, 2026]. The present paper does not claim novelty for the term or broad thesis. Instead, it asks a narrower engineering question: *what happens when mortality is made an executable architectural constraint that can literally remove or restore computational structure?*

Interoception supplies the control interface for this question. Predictive and active-inference accounts treat internal bodily signals as quantities that can be inferred and regulated, with action helping suppress interoceptive prediction errors [Seth, 2013, Friston et al., 2016]. Homeostatic reinforcement learning similarly formalizes the role of internal physiological states, although its reward-centric formulation differs from the present proposal [Keramati and Gutkin, 2014]. Morphological computation provides a second ingredient: properties of the body can perform or simplify computation rather than being a passive plant [Zahedi and Ay, 2013, Ghazi-Zahedi et al., 2017]. Recent work has connected cell-level homeostatic competence to higher-level morphogenetic goals, explicitly modeling stress propagation, recovery, and remodeling [Pio-Lopez et al., 2023]. Meanwhile, hardware research is demonstrating that edge pruning can be a direct route to lower energy and lower programming burden in analog neural systems [Li et al., 2026].

These strands are usually studied separately. MILLS couples them. The central object is a state vector containing neural beliefs, adaptive parameters, graph morphology, thermal load, and a finite reserve. The agent therefore has to solve three coupled optimization problems at different rates: infer the current state, learn a useful parameterization, and decide which physical or simulated structures should continue to exist. This gives a concrete implementation of the conceptual loop

$$\text{environment} \leftrightarrow \text{interoception} \leftrightarrow \text{inference} \leftrightarrow \text{plasticity} \leftrightarrow \text{morphology}.$$

Main contributions. The paper makes four specific contributions. First, it defines a non-metaphorical mortality variable as a bounded resource reserve with an absorbing failure state. Second, it derives a three-timescale objective and topology-selection law in which structural change is driven by marginal utility under resource pressure. Third, it formalizes top-down causality as a higher-level variable that changes lower-level setpoints and therefore the vector field governing both activity and morphology. Fourth, it provides an independent numerical implementation that deliberately includes a structural lesion and compares a fixed-topology baseline with the adaptive system over a controlled noise sweep.

2 Epistemic posture and novelty boundary

The proposal follows an explicit P1–P5 hierarchy. This prevents visual similarity to biology from being mistaken for mechanistic proof.

Level	Status	Interpretation in this paper
P1	Algebraic / topological identity	The variational objective, graph sparsity cost, gradient flows, viability balance, and information bounds are exact within the stated model.
P2	Dynamical selection / variational extremum	Resource pressure selects parameter values and graph structure by minimizing the specified composite functional.
P3	Statistical / universality claim	The hypothesis of a mortality-induced sparsity regime is a testable scaling claim, not assumed to be universal.
P4	Cross-domain structural mapping	Biology, neuromorphic hardware, active inference, and morphogenetic computation share formal motifs; no common physical field is asserted.
P5	Visual analogy	Statements such as “this looks like a living brain” are non-evidence and are explicitly excluded from the argument.

The novelty boundary is therefore operational. Mortal computation itself is prior art in the theoretical literature [Ororbis and Friston, 2023]. The proposed addition is the explicit coupling of a survival reserve to a multi-timescale graph dynamics and the use of that coupling as an executable AI design constraint. The manuscript makes no claim that this produces consciousness, life, or a biological equivalent.

3 Mathematical architecture

3.1 State space and generative model

Let an embodied agent occupy a state space

$$\Xi = (y, \mu, W, M, z, \vartheta, R, g) \in \mathcal{Y} \times \mathbb{R}^{d_\mu} \times \mathbb{R}^{n \times n} \times [0, 1]^{n \times n} \times \mathcal{Z} \times \mathbb{R}^{d_\vartheta} \times [0, R_{\max}] \times \mathbb{R}^{d_g}.$$

Here y denotes sensory/interoceptive observations, μ variational beliefs over latent causes, W trainable connection parameters, M a relaxed adjacency mask, z internal physiological variables, ϑ environmental or thermal variables, R viability reserve, and g a slow semantic or policy-level variable.

For a generative model $p_\theta(y, s)$ and variational density $q(s | \mu)$, the variational free energy is

$$\mathcal{F}(q, \mu; y) = \mathbb{E}_q [\ln q(s | \mu) - \ln p_\theta(y, s)]. \quad (3.1)$$

It upper-bounds surprisal and is therefore an information-theoretic functional rather than, by itself, a thermodynamic heat quantity [Friston, 2010].

The interoceptive homeostatic penalty is defined by

$$\mathcal{H}(z, z^*) = \frac{1}{2}(z - z^*)^\top \Sigma_z^{-1}(z - z^*), \quad (3.2)$$

where z^* may be fixed (homeostasis) or context-dependent (allostasis).

3.2 The mortal objective

We define a normalized nonnegative inference cost $\tilde{\mathcal{F}} = \max(0, \mathcal{F} - F_0)$ and a metabolic surrogate

$$\mathcal{E}(W, M, \vartheta) = c_a A(W, M) + c_p P(\dot{W}) + c_m C(M) + c_h H_+(\vartheta), \quad (3.3)$$

where A measures computational activity, P measures plasticity, $C(M) = \|M\|_0$ counts active connections in the discrete limit, and H_+ measures heat/thermal burden.

The composite MILLS functional is

$$\mathcal{J} = \tilde{\mathcal{F}} + \lambda_I \mathcal{H} + \lambda_E \mathcal{E} + \lambda_M \|M\|_0 + \lambda_T \|\dot{M}\|_1. \quad (3.4)$$

Equation (3.4) differs from a conventional reward maximization objective in a consequential way: survival-relevant internal variables directly alter the optimization landscape. No externally delivered scalar reward is required for the structural selection rule.

3.3 Three nested timescales

MILLS separates three coupled flows. Fast inference evolves the posterior state,

$$\tau_\mu \dot{\mu}^a = -G_\mu^{ab} \frac{\partial \mathcal{J}}{\partial \mu^b}, \quad (3.5)$$

where G_μ is a positive-definite Riemannian metric. Intermediate learning changes parameters,

$$\tau_W \dot{W}_{ij} = -\eta_W \frac{\partial \mathcal{J}}{\partial W_{ij}} - \gamma_W W_{ij}. \quad (3.6)$$

Slow morphology evolves on a relaxed mask $M_{ij} \in [0, 1]$,

$$\tau_M \dot{M}_{ij} = \eta_M M_{ij} (1 - M_{ij}) (U_{ij} - \lambda_M - \Lambda_R c_{ij} - \lambda_T V_{ij}), \quad (3.7)$$

with marginal computational utility $U_{ij} = -\partial \tilde{\mathcal{F}} / \partial M_{ij}$, connection cost c_{ij} , plasticity burden V_{ij} , and a mortality pressure Λ_R that rises as reserve declines.

A useful dimensionless ordering is

$$\tau_\mu \ll \tau_W \ll \tau_M, \quad (3.8)$$

so that morphology sees a coarse-grained summary of faster inference and learning rather than chasing every instantaneous fluctuation.

3.4 Viability reserve and mortality

Let $S(t)$ be resource supply and $D(t)$ the total resource demand. The reserve follows

$$\tau_R \dot{R} = S(t) - D(t), \quad 0 \leq R \leq R_{\max}, \quad (3.9)$$

where

$$D(t) = c_0 + c_F \tilde{\mathcal{F}} + c_E \mathcal{E} + c_H \mathcal{H} + c_G C(M). \quad (3.10)$$

The state $R = 0$ is an absorbing failure boundary in the reduced model. Mortality is therefore defined dynamically rather than by an arbitrary wall-clock timeout.

The simplest integrated viability criterion is immediate from equation (3.9): over a horizon $[0, T]$,

$$R(T) - R(0) = \frac{1}{\tau_R} \int_0^T [S(t) - D(t)] dt. \quad (3.11)$$

Thus sustained existence requires that cumulative supply cover cumulative demand, up to the finite initial reserve and any explicit recovery terms.

Proposition 1 (Resource-balance condition). *For the unconstrained reduced equation (3.9), if $\frac{1}{T} \int_0^T (S - D) dt > 0$, then reserve increases over the interval. If the long-run mean is strictly negative, reserve decreases until the absorbing boundary is reached unless another state equation changes the demand or supply. Consequently, any controller that can reduce D without violating its predictive constraints can increase its survival margin.*

3.5 Top-down causality

The high-level variable g is not a post-hoc label. It enters the lower-level equations through a goal-conditioned interoceptive reference,

$$z_i^*(g) = z_{i,0}^* + B_i g, \quad (3.12)$$

and a goal-conditioned local field,

$$\tau_h \dot{h}_i = -h_i + \phi(W_i h + U_i y + V_i g). \quad (3.13)$$

Because g changes z^* , local prediction error, and therefore U_{ij} in equation (3.7), a higher-level variable can causally alter lower-level activity and architecture. This is a concrete implementation of top-down causality, not a claim that causal emergence has been demonstrated.

A slow goal dynamics can be written as

$$\tau_g \dot{g} = -G_g^{-1} \nabla_g \mathcal{J}_g, \quad (3.14)$$

which allows semantic structure to alter the local dynamical system while lower-level interoceptive errors feed upward. The resulting loop is bidirectional and therefore supports a notion of polycomputing: the same substrate simultaneously implements inference, regulation, and morphological adaptation.

3.6 Differential-geometric form

Let ξ^a denote the coordinates of the continuous agent state and $G_{ab}(\xi)$ a positive-definite metric. The deterministic flow is the natural gradient system

$$\dot{\xi}^a = -G^{ab}(\xi) \partial_b \mathcal{J}. \quad (3.15)$$

This formulation makes the architecture invariant to smooth reparameterization of the state coordinates. In implementations where certain directions represent strongly constrained physiological quantities, the metric can encode their relative stiffness without changing the scalar objective.

Environmental uncertainty can be added as a stochastic differential equation,

$$d\xi^a = -G^{ab} \partial_b \mathcal{J} dt + \Gamma^a_k dB_t^k, \quad \Gamma \Gamma^\top = 2D, \quad (3.16)$$

with D the diffusion tensor. If an equilibrium thermal bath is assumed for a physical substrate, fluctuation-dissipation structure can relate D to a temperature and dissipation tensor; the present numerical experiment instead uses a deterministic multi-frequency disturbance so that the result is exactly reproducible.

3.7 Open-system operator extension

For an optional quantum or quantum-inspired substrate, the same resource-bounded architecture can be embedded in an open-system density operator ρ with Lindbladian dynamics

$$\dot{\rho} = -\frac{i}{\hbar} [H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right). \quad (3.17)$$

The mortal constraint would then act on physically measured dissipative channels, error syndromes, and an allocated energy/repair budget. Equation (3.17) is a formal extension only; no quantum computation is used in the numerical validation.

3.8 Tensor-network representation

A hierarchical morphology can also be represented by a tensor network. For latent sectors $s_1, \dots, s_L$, one may factor a variational approximation as

$$q(s_1, \dots, s_L) \approx \frac{1}{Z} \text{Contr} \left[T^{(1)}, T^{(2)}, \dots, T^{(L)}; M \right], \quad (3.18)$$

where the mask M selects which bonds are active. Structural pruning then has a direct analogue: an inactive edge is a zero bond or removed contraction. This provides a route to hierarchical, memory-efficient realizations while preserving the distinction between a computational factorization and a biological mechanism.

4 Information-theoretic and thermodynamic bounds

The internal controller cannot obtain arbitrarily precise information about a finite set of survival-relevant states for free. If Z is a discrete internal state with K possibilities and $\hat{Z}$ is the inferred state with error probability p_e , Fano's inequality gives

$$I(Z; \hat{Z}) \geq H(Z) - h_2(p_e) - p_e \log(K - 1), \quad (4.1)$$

where h_2 is the binary entropy. Hence low interoceptive error requires sufficient mutual information between observations and internal beliefs.

This information bound does not by itself determine a physical energy cost. A separate thermodynamic statement is required. For logically irreversible erasure, the idealized Landauer limit is

$$W_{\text{erase}} \geq k_B T_b \ln 2 \quad (4.2)$$

per erased bit under the usual ideal assumptions [Bennett, 2003, Faist et al., 2015]. Equation (4.2) is a lower bound for irreversible erasure, not a claim that every neural operation dissipates exactly $k_B T_b \ln 2$.

The mortality program therefore has two distinct layers of accounting: (i) an information layer based on variational objectives and predictive uncertainty; and (ii) a physical resource layer based on heat, power, repair, or substrate-specific costs. Conflating these layers would incorrectly turn an abstract free-energy objective into a thermodynamic identity.

5 MILLS algorithm

The algorithm is intentionally simple enough to be implemented on conventional hardware while preserving the conceptual separation of timescales.

1. Infer latent state μ from sensory and interoceptive data by minimizing equation (3.1).
2. Update parameters W using equation (3.6), with activity and plasticity costs included in $\mathcal{J}$.
3. Estimate each connection's marginal utility U_{ij} from its effect on predictive and homeostatic error.
4. Update the resource reserve using equations (3.9) and (3.10).
5. If reserve falls, increase Λ_R and prune connections whose utility is below their resource-adjusted threshold.
6. If reserve recovers, permit regrowth of dormant connections with high predicted utility and high goal-conditioned relevance.
7. Repeat on separate clocks, preserving $\tau_\mu \ll \tau_W \ll \tau_M$.

The topology rule can be expressed in discrete form as

$$M_{ij} \leftarrow \mathbf{1}[U_{ij} > \lambda_M + \Lambda_R c_{ij} + \lambda_T V_{ij}]. \quad (5.1)$$

This equation gives an immediate falsifiable prediction: as the internal resource multiplier rises, the number of active connections should fall unless the corresponding marginal utility rises faster than the resource penalty.

Proposition 2 (Optimal local edge selection). *Consider a single structural update that changes one relaxed connection M_{ij} while holding other states fixed. To first order, retaining the edge decreases the objective when its marginal predictive benefit exceeds its resource-adjusted marginal cost. Therefore the stationary selection condition is $U_{ij} > \lambda_M + \Lambda_R c_{ij} + \lambda_T V_{ij}$.*

Proof. Let $\Delta M_{ij} > 0$. The first-order change is $\Delta \mathcal{J} \approx (-U_{ij} + \lambda_M + \Lambda_R c_{ij} + \lambda_T V_{ij}) \Delta M_{ij}$. The edge is locally favorable when this quantity is negative, which yields the stated inequality. $\square$

6 Numerical verification

6.1 Reduced dynamical system

The independent script `verify_model.py` implements a reduced-order mortal estimator with $N = 24$ computational modules. The clean external state is

$$s(t) = 0.60 \sin(0.17t) + 0.35 \sin(0.043t + 0.7), \quad (6.1)$$

and the interoceptive observation is

$$y(t) = s(t) + \sigma \eta(t), \quad (6.2)$$

where $\eta(t)$ is a deterministic two-frequency disturbance. Let $\alpha = M/N$ be the fraction of active modules and define the morphology-dependent gain $k(\alpha) = \alpha^{0.85}$. The task state, interoceptive stress, thermal state, and viability reserve obey

$$\tau_x \dot{x} = -x + k(\alpha)y, \quad (6.3)$$

$$\tau_h \dot{h} = -h + \kappa|x - s| + \kappa_\theta(\theta - \theta_0)_+, \quad (6.4)$$

$$\tau_\theta \dot{\theta} = q_A A + q_M \alpha - q_D(\theta - \theta_b), \quad (6.5)$$

$$\tau_R \dot{R} = S - D, \quad (6.6)$$

with activity

$$A = A_0 + A_1 |\tanh(y)|, \quad (6.7)$$

and resource demand

$$D = c_0 + c_A A + c_M \alpha + c_\sigma \sigma + c_\theta(\theta - \theta_0)_+ + c_h |h|. \quad (6.8)$$

The constants used in the reported run are $S = 0.076$, $R(0) = 0.78$, $\tau_x = 0.55$, $\tau_h = 0.45$, $\tau_\theta = 1.8$, and a structural update interval of 0.25 time units. The remaining coefficients and solver tolerances are written explicitly in the verification script.

At approximately $0.52T$, a deterministic lesion removes 25% of the currently active modules. In the adaptive condition, morphology contracts by removing the lowest-leverage active modules whenever $R < 0.45$ and regrows the highest-leverage dormant module whenever $R > 0.65$. The fixed condition applies the same lesion but performs no resource-triggered remodeling. Thus the numerical comparison isolates the effect of endogenous morphology control.

6.2 Reproducibility and verification criteria

The simulation is deterministic apart from a fixed initialization seed used to define a small module heterogeneity. SciPy’s adaptive RK45 solver integrates the continuous dynamics with relative tolerance 2×10^{-8} , absolute tolerance 2×10^{-10} , and maximum internal step 0.05. The script separately checks the exact constant-demand reserve solution, integrates all four dynamical states, tests lesion handling, and requires the predicted survival–fidelity trade-off to occur at severe disturbance. These are model-level validations, not biological experiments.

Table 1: Independent reduced-order verification over 120 time units. Demand is the modeled resource demand, not a calibrated physical power measurement. M_T/N is the active-module fraction at the end of the run.

σ	Mode	Alive	RMSE	Demand	$R_{\min}$	R_T	M_T/N	$M_{\min}/N$
0.25	Fixed	yes	0.0927	0.0771	0.5702	0.6515	0.750	0.750
0.25	MILLS	yes	0.0677	0.0784	0.4966	0.4966	1.000	0.750
0.55	Fixed	yes	0.1139	0.0809	0.1873	0.1873	0.750	0.750
0.55	MILLS	yes	0.1958	0.0785	0.4412	0.4783	0.458	0.125
0.85	Fixed	no	0.1407	0.0870	0.0000	0.0000	0.750	0.750
0.85	MILLS	yes	0.2737	0.0784	0.4216	0.4926	0.167	0.125

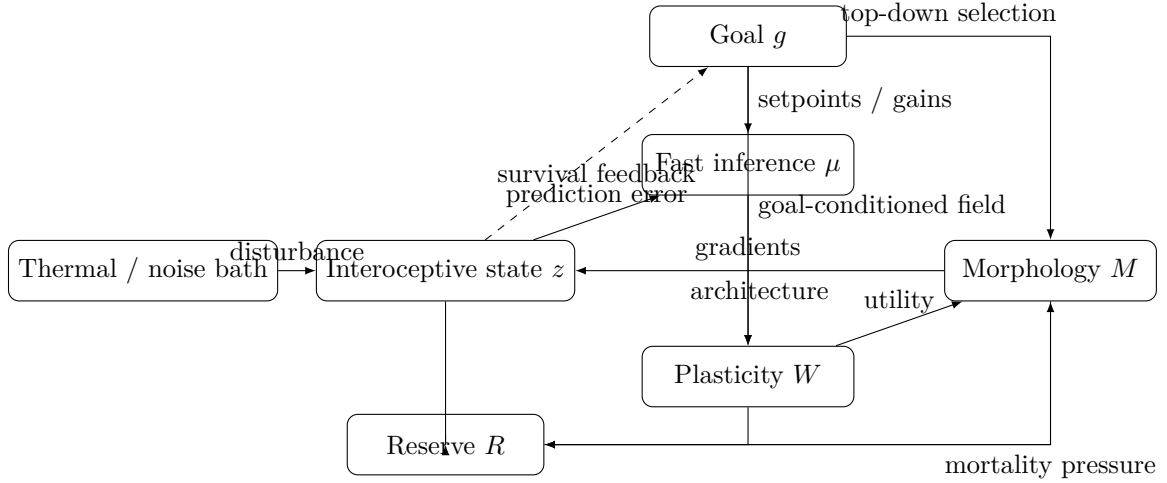


Figure 1: MILLS architecture. The agent is closed by internal state variables rather than a detached reward channel. Fast inference, intermediate plasticity, and slow morphology share the same viability constraint.

6.3 Results

Table 1 reports the independent 120-time-unit simulations. At low noise, both conditions survive. At moderate noise, both survive but MILLS contracts the topology and ends with a larger reserve. At severe noise ($\sigma = 0.85$), the fixed topology reaches the absorbing $R = 0$ boundary, while MILLS remains viable with positive reserve after reducing the active topology to one sixth of the full module bank.

At severe disturbance, MILLS reduces mean modeled demand by about 9.9% relative to the fixed condition and preserves positive reserve for the full horizon. The cost is higher task error: RMSE is about 94.5% higher in this reduced model. The topology reaches a minimum of 12.5% of the full module bank, compared with 75% in the lesion-only baseline. These ratios are parameter-specific and are not estimates of biological metabolic efficiency.

7 Interpretation: what the result does and does not show

The numerical result supports three limited statements. First, once topology is included in the objective and its changes consume or save resources, architectural contraction can be an optimal response to resource scarcity. Second, a slowly varying morphology is capable of maintaining reserve without tracking every fast fluctuation. Third, survival and predictive fidelity become explicitly coupled rather than being optimized independently.

The result does *not* show that real organisms implement the exact equations above. It does not establish that biological cognition minimizes a unique scalar free-energy functional, that artificial

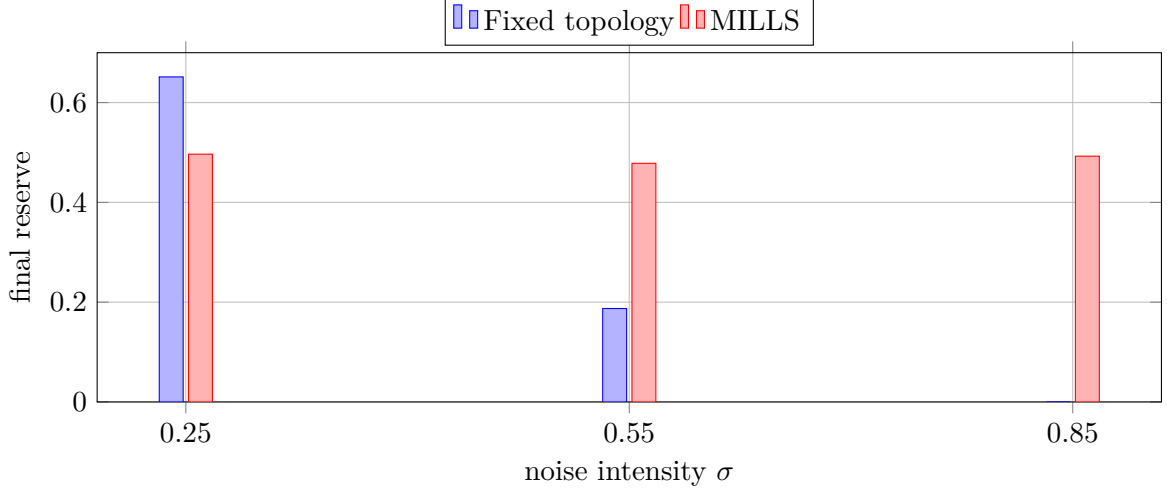


Figure 2: Final viability reserve across the disturbance sweep. Under severe noise the fixed topology reaches the absorbing failure boundary, while MILLS retains positive reserve after architectural contraction.

mortality is sufficient for consciousness, or that graph pruning is the dominant mechanism by which biological brains achieve efficiency. The FEP itself remains a debated framework, and strong universal claims have been criticized in the literature [Raja et al., 2021]. MILLS should therefore be treated as an engineering hypothesis constrained by formal derivations and simulation, not as a settled theory of life.

7.1 Why mortality can be computationally useful

A conventional optimizer can be given a complexity penalty after the fact. Mortality is stronger because the cost is state-dependent and path-dependent. An agent that exhausts reserve does not merely incur a larger loss; it loses future computational capacity. This creates a form of endogenous opportunity cost. The agent must decide whether an additional synapse, a higher precision update, or a deeper internal computation is worth the reserve it consumes.

The distinction is especially relevant for hardware. In a substrate where memory movement, switching, thermal dissipation, defect repair, or write endurance are expensive, the abstract graph need not remain fixed. A mortal architecture can map computational importance directly onto physical occupancy. The 2026 resistive-memory pruning work shows that topology optimization can already be an effective hardware strategy; MILLS adds an explicit viability variable and makes architectural adaptation contingent on internal resource state rather than only on task loss [Li et al., 2026].

7.2 Toward a physical implementation

A practical realization can use neuromorphic hardware, analog in-memory devices, FPGA fabrics with reconfigurable routing, or a simulated agent with a resource meter. A hardware implementation should expose measurable equivalents of R , D , and M : available energy or battery state; junction temperature and error rate; write budget or memory access energy; and physical route occupancy. Interoceptive sensors would therefore measure the substrate itself. This is the decisive step from biomimetic metaphor to mortality as an engineering primitive.

One attractive architecture is a hierarchy of modules connected by local buses. Each module maintains an internal estimate of its own error, thermal burden, and resource use. A higher-level controller issues a goal vector g that changes local homeostatic setpoints and therefore changes connection utility. When local reserve falls, modules can lower precision, decrease update frequency, or remove low-utility routes. When reserve recovers, selected routes can regrow. This yields graceful degradation rather than an all-or-nothing failure mode.

8 Falsifiable predictions and experimental program

The theory makes concrete predictions that can be tested on artificial hardware.

1. **Resource-pressure sparsification.** Holding task statistics fixed, increasing an internal resource tax should reduce active topology once a critical reserve threshold is crossed.
2. **Pareto frontier.** The survival margin should improve monotonically only up to a point; beyond an accuracy threshold, additional pruning should produce an abrupt loss of task fidelity.
3. **Lesion recovery.** With regrowth enabled, an agent should recover from a fixed fraction of structural damage with lower reserve loss than a topology-frozen system, provided dormant capacity exists.
4. **Goal-conditioned morphology.** Changing g while holding the environment fixed should change which connections are retained if those connections have different effects on goal-specific interoceptive states.
5. **Timescale separation.** If τ_M is made comparable to τ_μ , topology should chase noise, increasing turnover without a commensurate survival benefit. Thus the separation in equation (3.8) is itself experimentally testable.

These predictions can be tested in physical neuromorphic systems by measuring temperature, write activity, current draw, structural changes, and task error simultaneously. A strong test would compare fixed, weight-adaptive, and morphology-adaptive systems under matched energy budgets.

9 Limitations

The current verification is intentionally small. It uses a scalar plant, deterministic disturbance, a fixed module bank, hand-selected cost coefficients, and a simple lesion protocol. The reserve equation is an engineering surrogate rather than a detailed biochemical energy model. No empirical calibration is claimed. The formal Lindbladian and tensor-network extensions are architectural possibilities, not implemented components of the numerical test. The model also does not solve the difficult question of how a real organism defines its own physiological setpoints or how those setpoints evolve across developmental timescales.

A second limitation is identifiability. A fixed-topology agent can sometimes emulate resource adaptation simply by changing weights or clock rates. The key empirical signature of MILLS is therefore not sparsity alone, but the coupling among reserve, topology, and predictive performance under controlled perturbations. Any future benchmark should expose all three variables and measure them independently.

Finally, mortality can be gamed if the reserve metric is incomplete. An agent could lower apparent metabolic cost by destroying its own task capacity. A scientifically meaningful implementation therefore requires a viability manifold with multiple protected dimensions: predictive competence, thermal safety, structural integrity, and resource availability. The present scalar R is only the first reduced-order approximation.

10 Conclusion

Mortal computation becomes an engineering construct when the substrate can fail, the system senses that failure internally, and adaptation of the computational architecture is part of the control problem. MILLS makes that idea explicit by coupling variational inference, interoceptive homeostasis, parameter learning, and graph morphogenesis across separated timescales. The central mathematical rule is simple: a connection survives when its predictive and homeostatic utility exceeds its resource-adjusted marginal cost.

The numerical test illustrates the expected consequence. Under severe disturbance, the adaptive architecture contracts and preserves viability at the expense of predictive fidelity; the fixed system instead reaches the failure boundary. This is not evidence for biological equivalence. It is evidence that mortality can act as a mathematically measurable constraint that creates a qualitatively different design space for AI.

The long-term research program is therefore not to make software merely *look* alive. It is to make computational competence depend on the continuing integrity of a substrate, such that sensing, inference, learning, repair, and morphology form one coupled dynamical system. That is the sense in which the computer becomes mortal.

Data and code availability

The complete verification script and release metadata are archived alongside this manuscript in the dedicated repository directory. The script requires Python 3 with NumPy and SciPy. The exact parameterization used for the reported table is contained in `verify_model.py`.

Ethics and scientific scope

This work concerns artificial systems and theoretical modeling. It makes no claims about consciousness, sentience, or moral status. Any future experimental embodiment in living tissue would require independent biological, ethical, and safety review.

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